Partial Fractions and Improper Integrals

ANSWER KEY

Section 1: Linear Partial Fractions

1 Evaluate $\int \frac{1}{x^2-4}dx$.

a. $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$ ✓

b. $\frac{1}{2}\ln\left|\frac{x-2}{x+2}\right| + C$

c. $\frac{1}{4}\ln|x^2-4| + C$

d. $\frac{1}{2}\arctan\left(\frac{x}{2}\right) + C$

Decomposition: $\frac{1}{(x-2)(x+2)} = \frac{1/4}{x-2} - \frac{1/4}{x+2}$. Integrating gives $\frac{1}{4}\ln|x-2| - \frac{1}{4}\ln|x+2| + C = \frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$.

2 Evaluate $\int \frac{5x-3}{(x-1)(x+3)}dx$.

a. $\frac{1}{2}\ln|x-1| + \frac{9}{2}\ln|x+3| + C$ ✓

b. $2\ln|x-1| + 3\ln|x+3| + C$

c. $\ln|x-1| + 4\ln|x+3| + C$

d. $5\ln|x^2+2x-3| + C$

$5x-3 = A(x+3) + B(x-1)$. For $x=1$: $2 = 4A \Rightarrow A = \frac{1}{2}$. For $x=-3$: $-18 = -4B \Rightarrow B = \frac{9}{2}$.

3 When must long division be performed BEFORE applying partial fractions?

a. **When the degree of the numerator is greater than or equal to the degree of the denominator** ✓

b. When the denominator has repeated roots

c. When the numerator is a constant

d. Always

Partial fractions applies only to proper rational functions. If $\deg(P) \geq \deg(Q)$, polynomial long division must be performed first.

Section 2: Repeated Linear & Irreducible Factors

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4 What is the form of the partial fraction decomposition for $\frac{x+2}{x(x-1)^2}$?

a. $\frac{A}{x}+\frac{B}{x-1}+\frac{C}{(x-1)^2}$ ✓

b. $\frac{A}{x}+\frac{B}{(x-1)^2}$

c. $\frac{A}{x}+\frac{Bx+C}{(x-1)^2}$

d. $\frac{A}{x}+\frac{B}{x-1}$

For a repeated linear factor $(x-1)^2$, there must be a separate term for each increasing power up to the multiplicity: $\frac{B}{x-1}+\frac{C}{(x-1)^2}$.

5 Evaluate $\int \frac{1}{x^2+9}dx$.

a. $\frac{1}{3}\arctan(\frac{x}{3})+C$ ✓

b. $\arctan(\frac{x}{3})+C$

c. $\frac{1}{9}\ln(x^2+9)+C$

d. $\frac{1}{3}\ln|x+3|+C$

Formula: $\int \frac{1}{x^2+a^2}dx=\frac{1}{a}\arctan(\frac{x}{a})+C$. Here $a=3$, giving $\frac{1}{3}\arctan(\frac{x}{3})+C$.

6 Evaluate $\int_2^3 \frac{1}{x(x-1)}dx$.

a. $\ln(\frac{4}{3})$ ✓

b. $\ln 2$

c. $\ln 3$

d. $\frac{1}{6}$

$\frac{1}{x(x-1)}=\frac{1}{x-1}-\frac{1}{x}$. $[\ln|\frac{x-1}{x}|]_2^3=\ln(\frac{2}{3})-\ln(\frac{1}{2})=\ln(\frac{4}{3})$.

Section 3: Improper Integrals with Infinite Limits (Type 1)

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7 Evaluate $\int_1^\infty \frac{1}{x^2} dx$.

a. 1 ✓

b. ∞ (diverges)

c. 2

d. 0

$\lim_{b\to\infty} \int_1^b x^{-2} dx = \lim_{b\to\infty} [-\frac{1}{x}]_1^b = \lim_{b\to\infty} (1 - \frac{1}{b}) = 1.$

8 For what values of p does the improper integral $\int_1^\infty \frac{1}{x^p} dx$ converge?

a. $p > 1$ ✓

b. $p \geq 1$

c. $p < 1$

d. All real p

By the p-integral test, $\int_1^\infty \frac{1}{x^p} dx$ converges if and only if $p > 1$ (to $\frac{1}{p-1}$), and diverges for $p \leq 1$.

9 Evaluate $\int_0^\infty e^{-2x} dx$.

a. $\frac{1}{2}$ ✓

b. 1

c. ∞ (diverges)

d. 2

$\lim_{b\to\infty} [-\frac{1}{2} e^{-2x}]_0^b = 0 - (-\frac{1}{2}) = \frac{1}{2}.$

Section 4: Improper Integrals with Discontinuous Integrands (Type 2)

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10 Evaluate the improper integral $\int_0^1 \frac{1}{\sqrt{x}}dx$.

a. 2 ✓

b. 1

c. ∞ (diverges)

d. $\frac{1}{2}$

The integrand has an infinite discontinuity at $x = 0$: $\lim_{a \to 0^+} [2\sqrt{x}]_a^1 = 2(1) - 0 = 2$.

11 Evaluate $\int_0^1 \frac{1}{x}dx$.

a. **Diverges to** ∞ ✓

b. 1

c. 0

d. e

$\lim_{a \to 0^+} [\ln x]_a^1 = 0 - (-\infty) = \infty$. The integral diverges.

12 Evaluate $\int_{-1}^1 \frac{1}{x^2}dx$.

a. **Diverges** ✓

b. -2

c. 0

d. 2

The integrand has an interior vertical asymptote at $x = 0$. One must split: $\int_{-1}^0 \frac{1}{x^2}dx + \int_0^1 \frac{1}{x^2}dx$. Since $\int_0^1 x^{-2}dx = \infty$, the entire integral diverges.

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